

Poisson Distribution and the Poisson Process

Experiments yielding numerical values of a random variable X , the number of outcomes occurring during a given time interval or in a specified region, are called **Poisson experiments**.

The given time interval may be of any length, such as a minute, a day, a week, a month, or even a year.

Example: A Poisson experiment can generate observations for the random variable X representing the number of telephone calls received per hour by an office, the number of days school is closed due to snow during the winter, or the number of games postponed due to rain during a baseball season.

The specified region could be a line segment, an area, a volume, or perhaps a piece of material.

In such instances, X might represent the number of field mice per acre, the number of bacteria in a given culture, or the number of typing errors per page. A Poisson experiment is derived from the **Poisson process** and possesses the following properties.

- The number of outcomes occurring in one time interval or specified region of space is independent of the number that occur in any other disjoint time interval or region. In this sense we say that the Poisson process has no memory.
- The probability that a single outcome will occur during a very short time interval or in a small region is proportional to the length of the time interval or the size of the region and does not depend on the number of outcomes occurring outside this time interval or region.
- The probability that more than one outcome will occur in such a short time interval or fall in such a small region is negligible.

Poisson distribution A discrete random variable X is said to follow poisson distribution with parameter λ if its probability mass function is

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad x=0, 1, 2, \dots, \infty$$

Moment generating function

$$\begin{aligned} M_X(t) &= E(e^{tx}) = \sum_{x=0}^{\infty} e^{tx} P(X=x) = \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda} \lambda^x}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^t)^x}{x!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)} \end{aligned}$$

Mean $E(X) = \left[\frac{d}{dt} M_X(t) \right]_{t=0} = \left[e^{\lambda(e^t - 1)} \lambda e^t \right]_{t=0} = \lambda$

$$\therefore \boxed{\text{Mean} = \lambda}$$

$$\begin{aligned} E(X^2) &= \left[\frac{d^2}{dt^2} M_X(t) \right]_{t=0} = \left[\frac{d}{dt} \lambda e^t e^{\lambda(e^t - 1)} \right]_{t=0} \\ &= \lambda \left[e^t e^{\lambda(e^t - 1)} \lambda e^t + e^{\lambda(e^t - 1)} \cdot e^t \right]_{t=0} \\ &= \lambda [\lambda + 1] = \lambda^2 + \lambda \end{aligned}$$

$$\therefore \text{Var}(X) = E(X^2) - [E(X)]^2 = \lambda^2 + \lambda - \lambda^2$$

$$\boxed{\text{Var}(X) = \lambda}$$

Poisson Distribution: The probability distribution of the Poisson random variable X , representing the number of outcomes occurring in a given time interval.

The formula for the Poisson probability mass function is

$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, x = 0, 1, 2, 3, \dots$, where λ is the average number of outcomes. Mean = λ = variance; so that standard deviation = $\sqrt{\lambda}$.

Examples

- ① The number of blinds born in a town in a particular year.
- ② Number of mistakes committed in a typed page.
- ③ The number of students scoring very high marks in all subjects
- ④ The number of plane accidents in a particular week.
- ⑤ The number of defective screws in a box of 100, manufactured by a reputed company.
- ⑥ Number of suicides reported in a particular day

Conditions: Poisson distribution is the limiting case of binomial distribution under the following conditions:

- The number of trials n is indefinitely large i.e., $n \rightarrow \infty$.
- The probability of success p for each trial is very small; i.e., $p \rightarrow 0$.
- $np = \lambda$ (say) is finite, $\lambda > 0$.

Pb:1 Suppose on an average 1 house in 1000 in a certain district has a fire during a year. If there are 2000 houses in that district, what is the probability that exactly 5 houses will have a fire during the year?

Soln:-

Let X denote the number of houses have fire during the year.

$$p = \text{probability of fire accident during a year} = \frac{1}{1000} \\ = 0.001$$

$n =$ Total number of houses $= 2000$. (This experiment is the limiting case of binomial distribution)

$$\lambda = np = 2000 \times 0.001 = 2$$

$$P(X=5) = \frac{e^{-2} 2^5}{5!} = 0.036.$$

The number of monthly breakdowns of a computer is a random variable having a Poisson distribution with mean equal to 1.8. Find the probability that this computer will function for a month.

a) Without a breakdown

b) With only one breakdown and

c) With atleast one breakdown

Soln:- $X = \text{no. of monthly breakdowns of a computer}$

Given $\lambda = 1.8$

$$(a) P(X=0) = \frac{e^{-1.8} (1.8)^0}{0!} = e^{-1.8} = 0.1653$$

$$(b) P(X=1) = \frac{e^{-1.8} (1.8)^1}{1!} = e^{-1.8} (1.8) = 0.2975$$

$$(c) P(X \geq 1) = 1 - P(X=0) = 1 - e^{-1.8} = 0.8347$$

If the probability of a defective fuse from a manufacturing unit is 2%, in a box of 200 fuses, find the probability that a) exactly 4 fuses are defective b) more than 3 fuses are defective.

Soln:-

X = no. of defective fuses.

probability of a defective fuse (p) = 0.02

Total no. of fuses (n) = 200.

$$\therefore \lambda = n \cdot p = 4$$

$$\therefore P(X=x) = \frac{e^{-4} 4^x}{x!} \quad x = 0, 1, 2, \dots, \infty$$

$$(a) \quad P(X=4) = \frac{e^{-4} 4^4}{4!} = 0.1952$$

$$\begin{aligned} (b) \quad P(X > 3) &= 1 - P(X \leq 3) \\ &= 1 - [P(X=0) + P(X=1) + P(X=2) + P(X=3)] \\ &= 1 - e^{-4} \left[1 + \frac{4}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} \right] \\ &= 0.5669 \end{aligned}$$

Fit a Poisson distribution to the following data and calculate the theoretical frequencies.

Deaths	0	1	2	3	4
Frequency	122	60	15	2	1